

Special Problem Defense

# MitoChime

A Machine Learning Pipeline for Detecting PCR-Induced  
Chimeric Reads in Mitochondrial Illumina Data

---

Duranne Duran · Yvonne Lin · Daniella Pailden

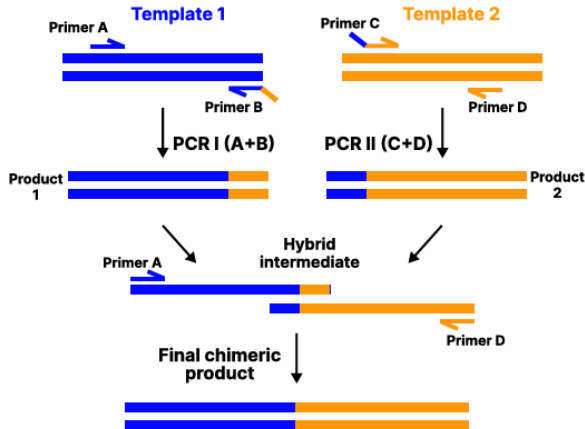
University of the Philippines Visayas  
Division of Physical Sciences and Mathematics

## Four simulated *Sardinella lemuru* mtDNA short reads

|        |                                                                |
|--------|----------------------------------------------------------------|
| Read 1 | TCAGCCTTTACCAAGGCCCCACCCTTGCGGCCCTCGCCTGCTTAATTTTACTTTCTTA...  |
| Read 2 | TAAGTAAGAACTAGTAGGATCGAGGCCTTCCCGTCTAGATAGGCGTTGGCTTAATTGCA... |
| Read 3 | ATAGGGCAGAAGGCCTAGCATGTTAATGGTAAATAGGAAAACCATTAGAGACGTAAATG... |
| Read 4 | GAGGTGGCGAAAGTCAGCGTGGTTCCCAGTCCAAGGCTCATCAGCAAGACGGTTAGGCT... |

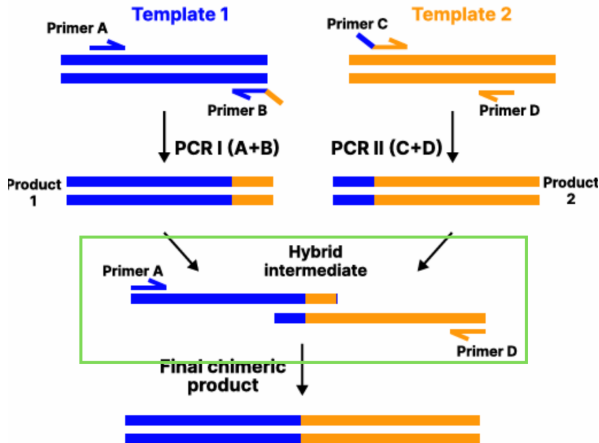
**Question:** Out of these short reads, which ones are chimeric?

# The Problem



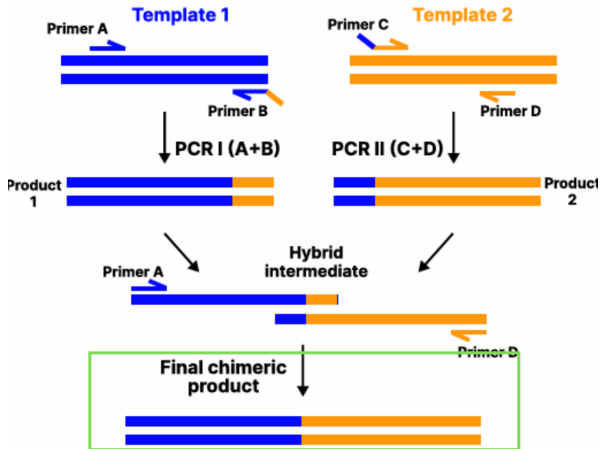
*Schematic illustration of PCR-induced chimera formation through incomplete extension and template switching.*

# The Problem



*Schematic illustration of PCR-induced chimera formation through incomplete extension and template switching.*

# The Problem



*Schematic illustration of PCR-induced chimera formation through incomplete extension and template switching.*

# Implications

## Assembly quality

- ▶ Fragmented or failed mitochondrial assemblies
- ▶ Reduced reliability of mitochondrial genomics workflows

## Research impact

- ▶ Limited downstream analyses in fisheries, population genetics, and evolution
- ▶ Wasted sequencing effort, computing time, and researcher labor

# Existing approaches

| Tool / Approach  | Main focus                       | Gap for this study                    |
|------------------|----------------------------------|---------------------------------------|
| UCHIME / UCHIME2 | Amplicon chimeras                | Reference/abundance dependent         |
| DADA2            | ASV-level filtering              | Not read-level mtDNA filtering        |
| CATCh            | 16S rRNA ensemble ML             | Not for mtDNA; inconsistent           |
| ChimPipe         | RNA-seq fusion/chimera detection | Not PCR-induced mtDNA artifacts       |
| DL-based tools   | DNA sequence motif prediction    | Few to no tools for chimera detection |
| <b>MitoChime</b> | <b>Read-level mtDNA chimeras</b> | <b>Pre-assembly ML/DL filtering</b>   |

Most existing tools target amplicon, ASV, or RNA-seq chimeras; MitoChime targets PCR-induced mitochondrial short-read artifacts.

Can machine learning detect PCR-induced chimeric reads  
**before mitochondrial genome assembly?**

### Input

Simulated clean and chimeric paired-end  
Illumina reads from *S. lemuru*

### Output

A modular pipeline that predicts and filters  
suspected chimeric reads



# Scope and limitations

## Included

- ▶ *Sardinella lemuru*
- ▶ mitochondrial Illumina reads
- ▶ PCR-induced chimeras
- ▶ simulated labelled datasets

## Excluded

- ▶ long-read sequencing
- ▶ NUMTs
- ▶ naturally occurring chimeras
- ▶ multi-species model training

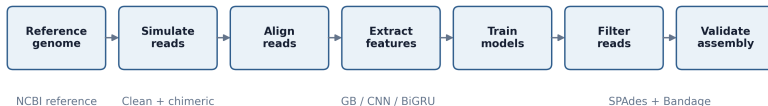
The study prioritized a focused and controlled evaluation setting.

# Methodology

From simulated reads to model-based filtering

# MitoChime workflow

## MitoChime workflow



## Overview

The workflow generated labelled reads, extracted read-level evidence, trained models, filtered suspected chimeras, and evaluated downstream assembly behavior.

# Data construction

## Clean reads

Simulated from the *S. lemuru* mitochondrial reference genome.

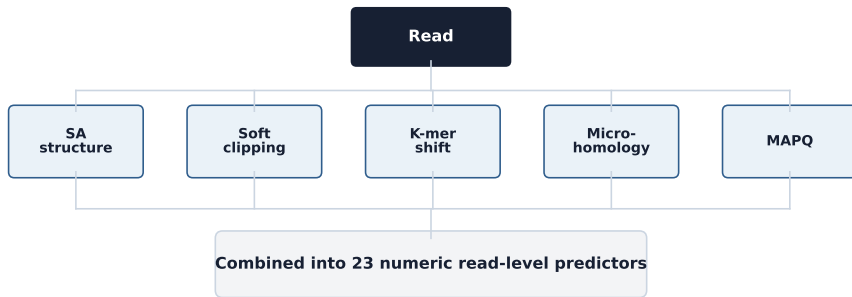
## Chimeric reads

Generated by joining non-adjacent mitochondrial segments to mimic PCR template switching.

Labels: **clean** = 0    and    **chimeric** = 1

# Feature extraction

## Read-level feature groups



## Feature representation

Each read was converted into a set of 23 numeric predictors describing alignment structure, clipping behavior, sequence composition, microhomology, and mapping quality.

# Dataset representation

The labelled reads were prepared in two dataset formats.

| read_id                                     | label | seq                                         |
|---------------------------------------------|-------|---------------------------------------------|
| NC_039553.1_10001_10492_5:0:0_8:0:0_b77/1   | 0     | GTAAGGCATAGTAAGGCAATGGTGAGGCTGTGTGTGATTGCA  |
| NC_039553.1_10001_10492_5:0:0_8:0:0_b77/2   | 0     | GAGTGAGCCGAATAGGGAATTAGTCCAACCTAAGACTCCTGA  |
| NC_039553.1_10003_10548_5:0:0_8:0:0_f4d/1   | 0     | GTGAGCCGAAAAGGTAATTAGTCCAACCTACGACTCCTGATT  |
| NC_039553.1_10003_10548_5:0:0_8:0:0_f4d/2   | 0     | TGGCTATATAGGTGTTGGTGAGGGCTCAGGCCGTTTCCGTTG  |
| NC_039553.1_10005_10512_6:0:0_6:0:0_c35/1   | 0     | CCGTTGTCCAGTTTAGCCAGGTAAGGCTTAGTAAGGCAATGC  |
| NC_039553.1_10005_10512_6:0:0_6:0:0_c35/2   | 0     | GAGCCGCATAGGGCAATAGTCCAACCTAAGACTCGTGATTTC  |
| NC_039553.1_10008_10523_3:0:0_10:0:0_914/1  | 0     | TCAGCCCCTTTCCGTTGACCAGTTTAGCCAGGTAAGGCTTAC  |
| NC_039553.1_10008_10523_3:0:0_10:0:0_914/2  | 0     | CCGAATAGGGACATAGTCCAACCTAAGACTCCTGATTCTGCG  |
| NC_039553.1_10012_10513_12:0:0_5:0:0_1340/1 | 0     | TCCGTTGTGCGAGTTTAGCCAGGTAAGGCTTAGTAAGGCAATG |
| NC_039553.1_10012_10513_12:0:0_5:0:0_1340/2 | 0     | ATAGTGAATTAGTCCAAGTTAAGACTCCTGATTTCGGCTCAGA |
| NC_039553.1_10012_10591_3:0:0_5:0:0_2571/1  | 0     | AGCCAGCATGTAAGAACCACAGCGGGGAGGATCAGGGATC    |
| NC_039553.1_10012_10591_3:0:0_5:0:0_2571/2  | 0     | ATATGGAATTAGTCCAACCTAAGACTCCTGATTTCGGCTCAGA |
| NC_039553.1_10015_10459_2:0:0_8:0:0_253e/1  | 0     | GTGATTGCAGCAGTTCAAAGTGACTTTTATAGCGCGAAATCAG |
| NC_039553.1_10015_10459_2:0:0_8:0:0_253e/2  | 0     | GGGAATTAGTCCAACCTAAGAGTCTGATTTCCGGCTCAGATG  |
| NC_039553.1_10015_10585_10:0:0_11:0:0_d4c/1 | 0     | GGGAATTAGACGAACATAATCTCCTGAATTCGGCTCAGATG   |
| NC_039553.1_10015_10585_10:0:0_11:0:0_d4c/2 | 0     | CATGTAAGAACCCACAGCGGGGAGGCTAAGGGATCCCTGGG   |

| read_id                               | label | read_len | mean_bp | ref_name    | ref_strand | mapq | cigar    | has_si | sa_count | num_seg | sa_diff | config | sa_min_delta_pos | sa_max_delta_pos |
|---------------------------------------|-------|----------|---------|-------------|------------|------|----------|--------|----------|---------|---------|--------|------------------|------------------|
| NC_039553.1_3_540_8:0:0_6:0:0_a12     | 0     | 100      | 13.000  | NC_039553.1 | 3          | 0    | 60 150M  | 0      | 0        | 1       | 0       |        | 0                | 0                |
| NC_039553.1_4_430_13:0:0_11:0:0_243d  | 0     | 150      | 13.000  | NC_039553.1 | 4          | 0    | 60 150M  | 0      | 0        | 1       | 0       |        | 0                | 0                |
| NC_039553.1_5_495_8:0:0_11:0:0_1440   | 0     | 150      | 13.000  | NC_039553.1 | 5          | 0    | 60 150M  | 0      | 0        | 1       | 0       |        | 0                | 0                |
| NC_039553.1_6_523_8:0:0_9:0:0_82u     | 0     | 150      | 13.000  | NC_039553.1 | 6          | 0    | 60 150M  | 0      | 0        | 1       | 0       |        | 0                | 0                |
| NC_039553.1_9_574_7:0:0_7:0:0_181b    | 0     | 150      | 13.000  | NC_039553.1 | 9          | 0    | 60 150M  | 0      | 0        | 1       | 0       |        | 0                | 0                |
| NC_039553.1_10_591_9:0:0_8:0:0_256b   | 0     | 150      | 13.000  | NC_039553.1 | 10         | 0    | 60 150M  | 0      | 0        | 1       | 0       |        | 0                | 0                |
| NC_039553.1_11_508_8:0:0_11:0:0_a19   | 0     | 150      | 13.000  | NC_039553.1 | 11         | 0    | 60 150M  | 0      | 0        | 1       | 0       |        | 0                | 0                |
| NC_039553.1_12_427_9:0:0_9:0:0_157    | 0     | 150      | 13.000  | NC_039553.1 | 12         | 0    | 60 150M  | 0      | 0        | 1       | 0       |        | 0                | 0                |
| NC_039553.1_12_489_8:0:0_9:0:0_86f    | 0     | 150      | 13.000  | NC_039553.1 | 12         | 0    | 60 150M  | 0      | 0        | 1       | 0       |        | 0                | 0                |
| NC_039553.1_12_593_13:0:0_13:0:0_2375 | 0     | 150      | 13.000  | NC_039553.1 | 12         | 0    | 24 150M  | 0      | 0        | 1       | 0       |        | 0                | 0                |
| NC_039553.1_14_515_7:0:0_7:0:0_20a4   | 0     | 150      | 13.000  | NC_039553.1 | 14         | 0    | 60 150M  | 0      | 0        | 1       | 0       |        | 0                | 0                |
| NC_039553.1_15_525_9:0:0_9:0:0_1e76   | 0     | 150      | 13.000  | NC_039553.1 | 15         | 0    | 60 150M  | 0      | 0        | 1       | 0       |        | 0                | 0                |
| NC_039553.1_17_520_4:0:0_8:0:0_25c    | 0     | 150      | 13.000  | NC_039553.1 | 17         | 0    | 60 148MM | 0      | 0        | 1       | 0       |        | 0                | 0                |
| NC_039553.1_18_522_8:0:0_9:0:0_c84    | 0     | 150      | 13.000  | NC_039553.1 | 18         | 0    | 60 150M  | 0      | 0        | 1       | 0       |        | 0                | 0                |
| NC_039553.1_18_593_8:0:0_9:0:0_sca    | 0     | 150      | 13.000  | NC_039553.1 | 18         | 0    | 60 150M  | 0      | 0        | 1       | 0       |        | 0                | 0                |
| NC_039553.1_18_482_8:0:0_7:0:0_1e4b   | 0     | 150      | 13.000  | NC_039553.1 | 18         | 0    | 60 150M  | 0      | 0        | 1       | 0       |        | 0                | 0                |
| NC_039553.1_19_516_8:0:0_8:0:0_c1a    | 0     | 150      | 13.000  | NC_039553.1 | 19         | 0    | 60 150M  | 0      | 0        | 1       | 0       |        | 0                | 0                |
| NC_039553.1_20_525_9:0:0_10:0:0_1b37  | 0     | 150      | 13.000  | NC_039553.1 | 20         | 0    | 60 150M  | 0      | 0        | 1       | 0       |        | 0                | 0                |
| NC_039553.1_21_560_10:0:0_9:0:0_189b  | 0     | 150      | 13.000  | NC_039553.1 | 21         | 0    | 60 150M  | 0      | 0        | 1       | 0       |        | 0                | 0                |
| NC_039553.1_23_443_2:0:0_8:0:0_1009   | 0     | 150      | 13.000  | NC_039553.1 | 23         | 0    | 60 150M  | 0      | 0        | 1       | 0       |        | 0                | 0                |
| NC_039553.1_25_571_7:0:0_8:0:0_1932   | 0     | 150      | 13.000  | NC_039553.1 | 25         | 0    | 60 150M  | 0      | 0        | 1       | 0       |        | 0                | 0                |
| NC_039553.1_28_536_8:0:0_8:0:0_182    | 0     | 150      | 13.000  | NC_039553.1 | 28         | 0    | 60 150M  | 0      | 0        | 1       | 0       |        | 0                | 0                |
| NC_039553.1_32_487_8:0:0_8:0:0_2005   | 0     | 150      | 13.000  | NC_039553.1 | 32         | 0    | 60 150M  | 0      | 0        | 1       | 0       |        | 0                | 0                |
| NC_039553.1_34_520_8:0:0_9:0:0_691    | 0     | 150      | 13.000  | NC_039553.1 | 34         | 0    | 60 150M  | 0      | 0        | 1       | 0       |        | 0                | 0                |
| NC_039553.1_34_452_2:0:0_16:0:0_495   | 0     | 150      | 13.000  | NC_039553.1 | 34         | 0    | 60 150M  | 0      | 0        | 1       | 0       |        | 0                | 0                |
| NC_039553.1_36_574_7:0:0_8:0:0_993c   | 0     | 150      | 13.000  | NC_039553.1 | 36         | 0    | 60 150M  | 0      | 0        | 1       | 0       |        | 0                | 0                |

## Feature matrix

For machine learning models

## Sequence dataset

For sequence-based deep learning models

# Models evaluated

## Gradient Boosting

Feature-based model using engineered alignment and sequence-derived predictors.

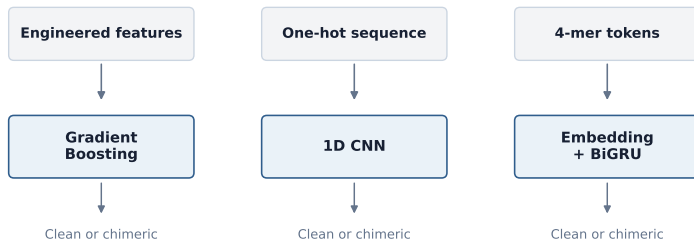
## 1D CNN

Sequence-based model using one-hot encoded nucleotide reads.

## BiGRU

Sequence-based model using ordered k-mer tokens.

### Three model representations



# Evaluation strategy

## Read-level evaluation

- ▶ Accuracy
- ▶ Precision
- ▶ Recall
- ▶ F1-score
- ▶ ROC-AUC

## Assembly-level validation

- ▶ Read retention
- ▶ Number of nodes
- ▶ Assembly graph structure
- ▶ Contiguity after filtering

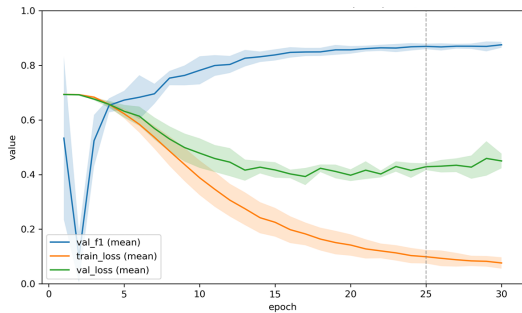
The study evaluated both prediction quality and downstream utility.



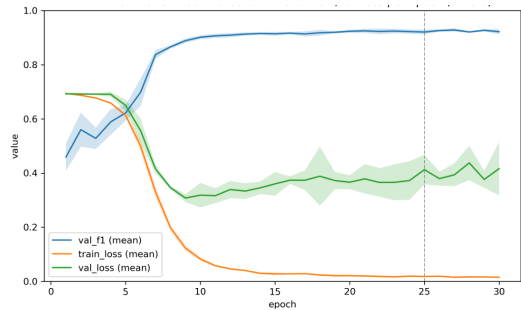
# Results and Discussion

Classification performance and assembly behavior

# Training Behavior of CNN and BiGRU

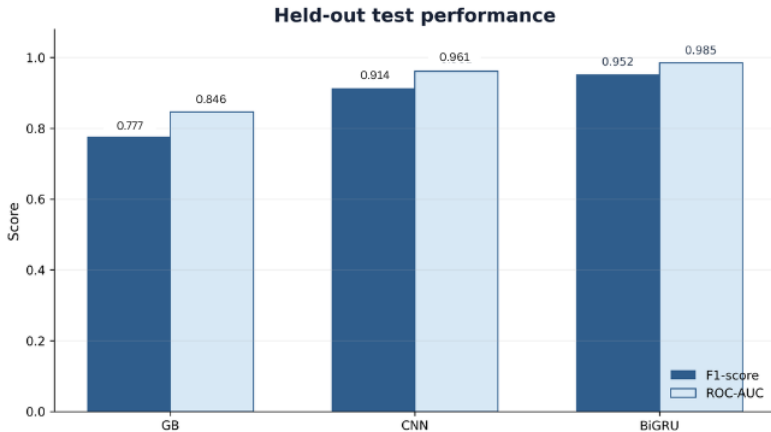


CNN



BiGRU

# Classification performance

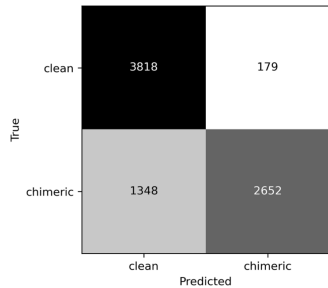


# Classification performance

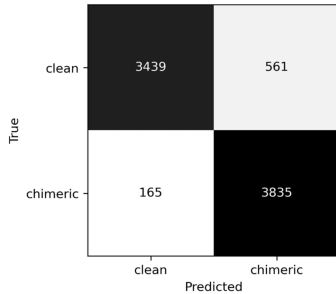
| Model             | F1-score | ROC-AUC | Runtime  |
|-------------------|----------|---------|----------|
| Gradient Boosting | 0.7765   | 0.8459  | 0.50 min |
| 1D CNN            | 0.9135   | 0.9607  | 1.39 min |
| BiGRU             | 0.9523   | 0.9849  | 5.51 min |

- ▶ **Gradient Boosting** — Conservative filtering; higher read retention
- ▶ **CNN** — Sensitive model; more aggressive filtering
- ▶ **BiGRU** — Highest F1 score; best read-level performance

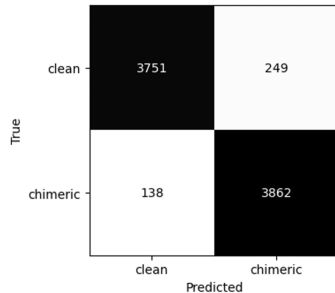
# Error patterns



Gradient Boosting

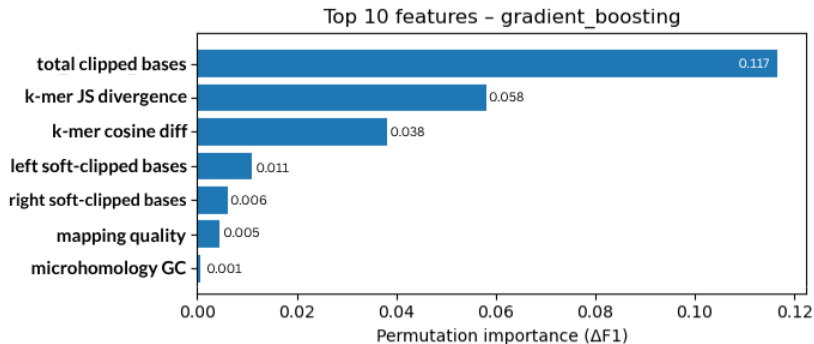


CNN



BiGRU

# Feature importance



Most informative predictors: **total clipped bases**, **k-mer Jensen–Shannon divergence**, **k-mer cosine difference**, and **left soft clipping**.

# Feature Family Importance

| Family        | Importance Sum | Importance Mean | nFeatures |
|---------------|----------------|-----------------|-----------|
| Clipping      | 0.133830       | 0.033458        | 4         |
| K-mer jump    | 0.096264       | 0.048132        | 2         |
| Other         | 0.003676       | 0.001225        | 3         |
| Microhomology | 0.000347       | 0.000174        | 2         |
| SA structure  | -0.000859      | -0.000066       | 13        |



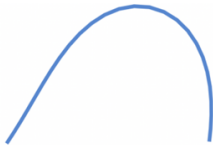
# Assembly validation: high coverage



(a) Unfiltered



(b) GB



(c) CNN



(d) BiGRU

20K reads, 50% chimeras

- ▶ Unfiltered assembly showed severe graph complexity.
- ▶ Filtering simplified the graph.
- ▶ CNN produced the simplest graph.
- ▶ GB retained more reads.

# Assembly validation: low coverage

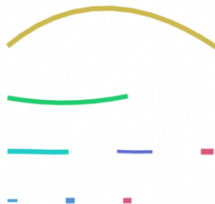


(a) Unfiltered

(b) GB



(c) CNN



(d) BiGRU

3,200 reads, 50% chimeras

At low read depth, aggressive filtering may remove too much read support and cause fragmentation.

## Practical implication

The safest model depends on coverage and chimera proportion.

## Recommendations: Practical Use

- ▶ Use **Gradient Boosting** when read depth is low.
- ▶ Use **BiGRU** when read-level classification is the priority.
- ▶ **CNN** may be useful in high-coverage and high-chimera conditions.

## Recommendations: Future Work

- ▶ Test on real mitochondrial datasets.
- ▶ Evaluate realistic class imbalance.
- ▶ Add varied sequencing-error profiles.
- ▶ Explore hybrid feature-sequence models.

# Summary

- ① PCR-induced chimeras can disrupt mitochondrial genome assembly.
- ② Existing tools do not directly address this read-level setting.
- ③ MitoChime detects suspected chimeric reads before assembly.
- ④ BiGRU achieved the strongest classification performance.
- ⑤ Model choice should consider both accuracy and read retention.

# MitoChime

A Machine Learning Pipeline for Detecting PCR-Induced  
Chimeric Reads in Mitochondrial Illumina Data

---

Duranne Duran · Yvonne Lin · Daniella Pailden

Thank you.